BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: Mathematics Paper–IX (01) (22)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** Mathematics Paper–IX (01) (22)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. For A = [[5, 6], [−4, 1]], |A| = ...

(A) 29 (B) −29

(C) −19 (D) 19

Q2. Subtracting 2 + 3i from 6 + 3i gives.....

(A) −4 (B) −4 − 6i

(C) 4 (D) 4 + 6i

Q3. √0 = ...

(A) 0 (B) −1

(C) 1 (D) Not defined

Q4. 4^{-3} = ...

(A) 64 (B) −64

(C) 1/64 (D) −1/64

Q5. Base of natural logarithm is.....

(A) 1 (B) 10

(C) π (D) e

Q6. $(a + b)(a^2 - ab + b^2) =$ .....

(A) $a^3 + b^3$ (B) $a^3 - b^3$

(C) $(a + b)^3$ (D) $(a + b)(a^2 - b^2)$

Q7. In 7/2, the remainder is.....

(A) 1 (B) 2

(C) 3 (D) 7

Q8. $x^2 - 5x + 6 =$ .....

(A) $(x-2)(x+3)$ (B) $(x+2)(x-3)$

(C) $(x-2)(x-3)$ (D) $(x+2)(x+3)$

Q9. Which one is the solution of $|-x| = 0$?

(A) $\{1\}$ (B) $\{-1\}$

(C) $\{0\}$ (D) $\{\}$

Q10. The line $y = 5$ is parallel to.....

(A) x−axis (B) y−axis

(C) Both A and B (D) Neither A nor B

Q11. In ΔPQR, if $m\angle P = m\angle Q$, then it is .....triangle.

(A) Equilateral (B) Scalene

(C) Isosceles (D) Equiangular

Q12. Which of the given are Concurrent?

(A) Angle bisectors (B) Medians

(C) Both A and B (D) Neither A nor B

Q13. The sum of two sides of a triangle is always.....the length of the third side.

(A) Equal to (B) Lesser than

(C) Greater than (D) Congruent to

Q14. If perpendicular distance between the lines is same, then they are....

(A) Perpendicular (B) Parallel

(C) Congruent (D) Intersecting

Q15. Area of a triangle with base 4 units and altitude 6 units, is...

(A) 6 square units (B) 10 square units

(C) 12 square units (D) 24 square units

Section-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 36

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

QII. Attempt any NINE parts. Each part carries FOUR marks.

1. For $X = [[1, 2], [2, 3]]$, $Y = [[1, 0], [0, -1]]$ evaluate $3X - 2Y$
2. Solve without using calculator, $\log_x(0.001) = -3$
3. Divide $(x^2 + x - 2)/(3x^2 + 9x + 6)$ by $x - 1$
4. If $u - v = 3$ then prove that $u^3 - v^3 - 9uv = 27$
5. Using long division, find the quotient and remainder when $x^3 + 2x^2 - 3x + 1$ is divided by $x + 2$
6. Find the multiplicative inverse of $M = [[1, 0], [0, 1]]$
7. Evaluate $(2x - 3)/(x^2 - x + 1)$ for $x = 2$
8. Factorize $25a^2b^2 - 20abc + 4c^2 - 16d^2$
9. Solve $(8x + 1)/2 < 2x$; $x \in R$
10. Draw a parallelogram by joining the points A(0,0), B(1,4), C(4,2), D(3,−2)
11. Find the square root of $4x^4 - 4x^3 + 13x^2 - 6x + 9$ by division method.
12. If $x + 1/x = 3$, evaluate $x^3 + 1/x^3$

Section-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 24

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII. Prove that any point on the bisector of an angle is equidistant from its arms.

QIV. Prove that from a point outside a line, the perpendicular is the shortest distance from the point to the line.

QV. If a line segment intersects the two sides of a triangle in the same ratio, then it is parallel to the third side.

QVI. Construct a rectangle ABCD, with adjacent sides 2.5 cm and 5 cm respectively.